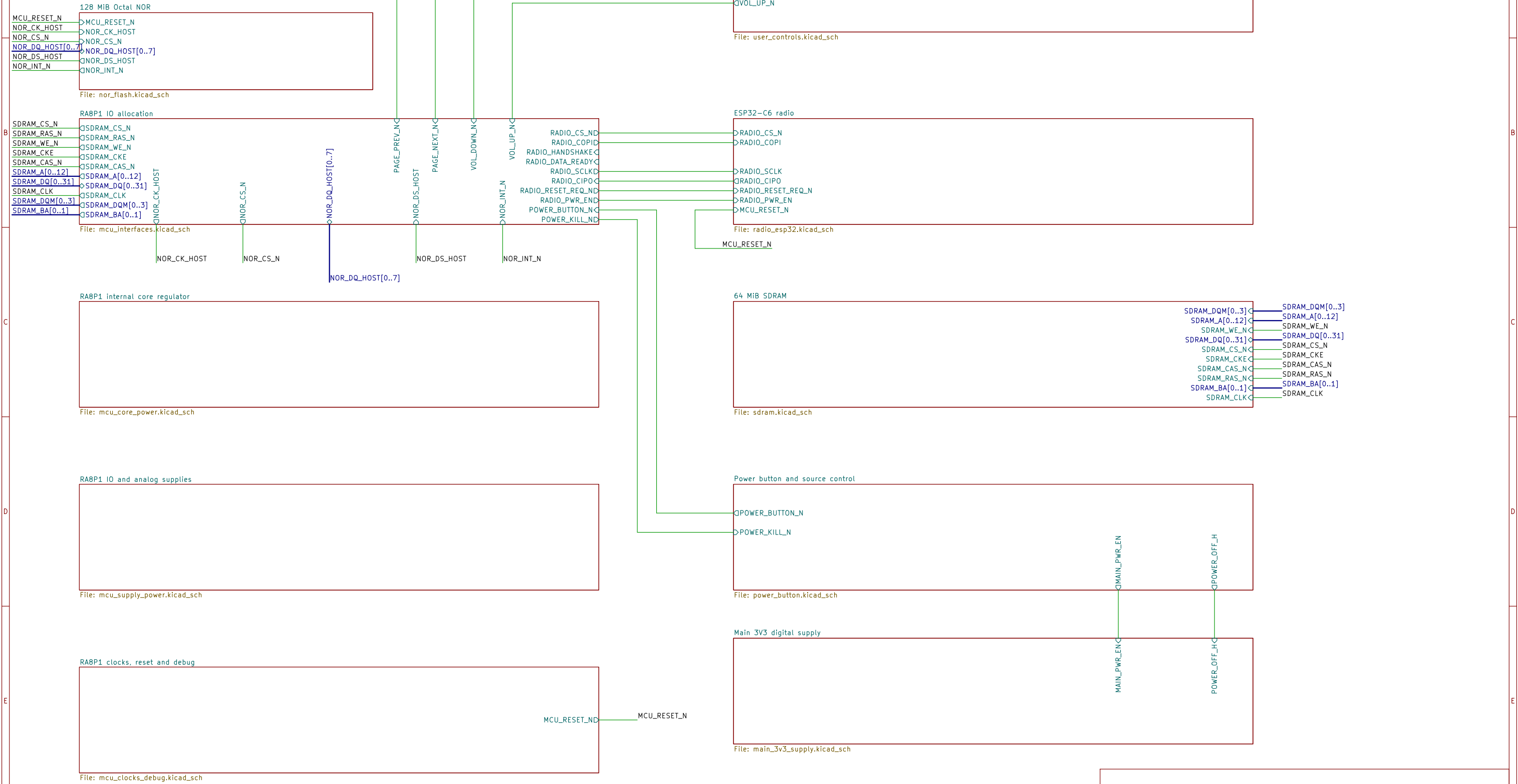


RA8P1 E-READER

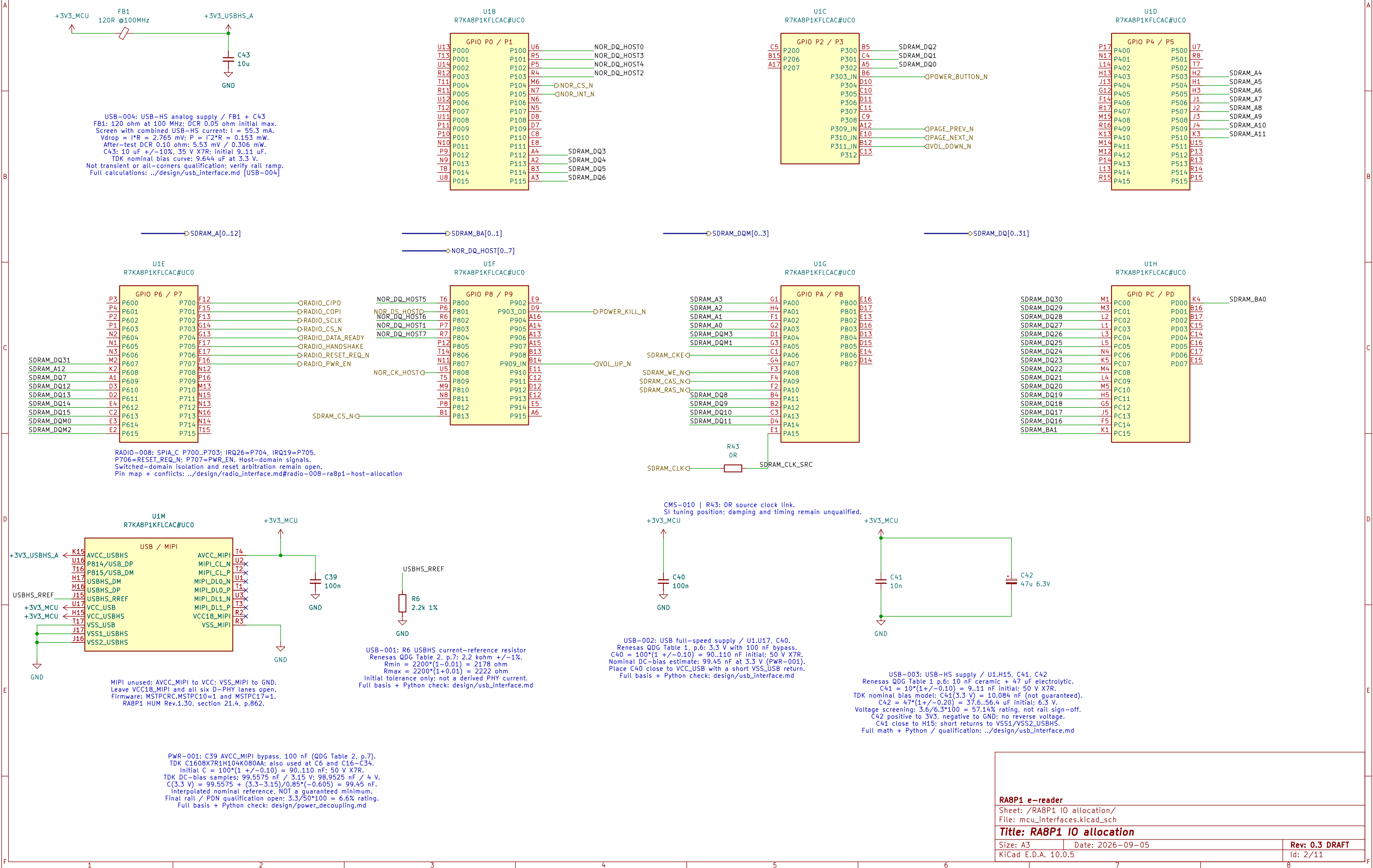
Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

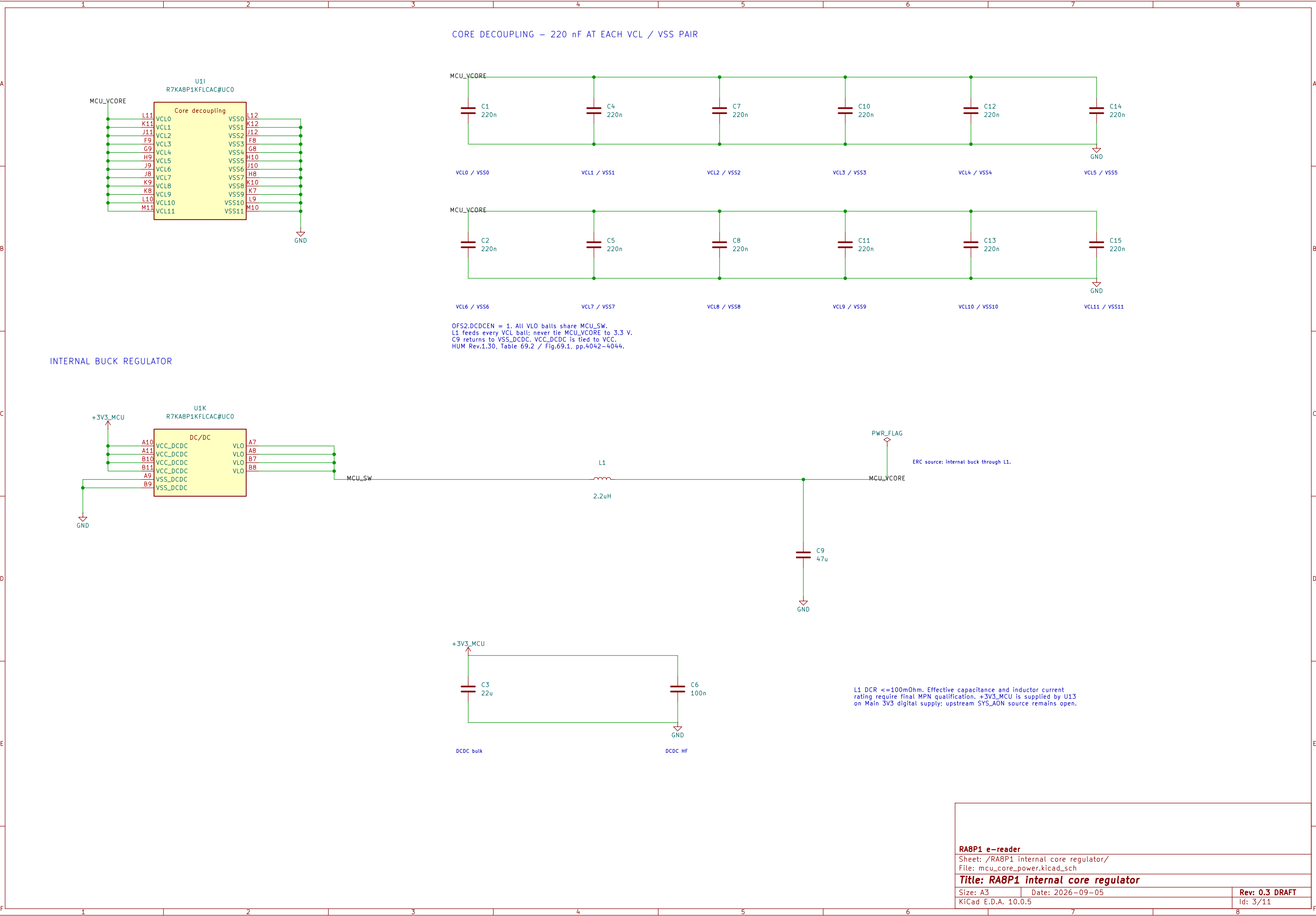


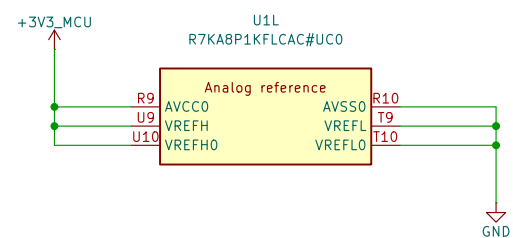
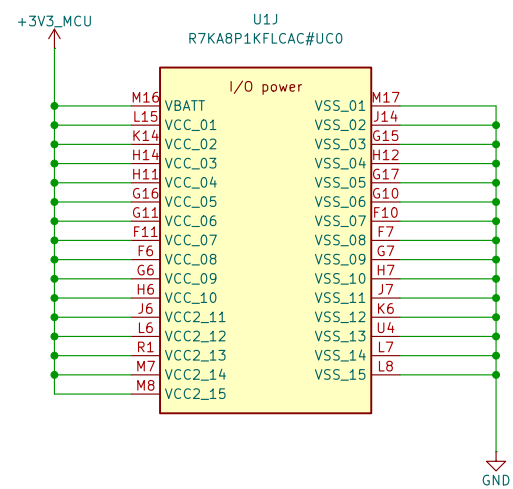
Global supplies: +3V3_MCU and GND.
MCU_VCORE remains local to the processor core-power sheet.
Circuit notes link to the supporting engineering calculations.
Footprint and PCB qualification are deferred by scope.

Sheet: /		
File: ereader_rev1.kicad_sch		
Title: RA8P1 e-reader – schematic index		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.5		Id: 1/11

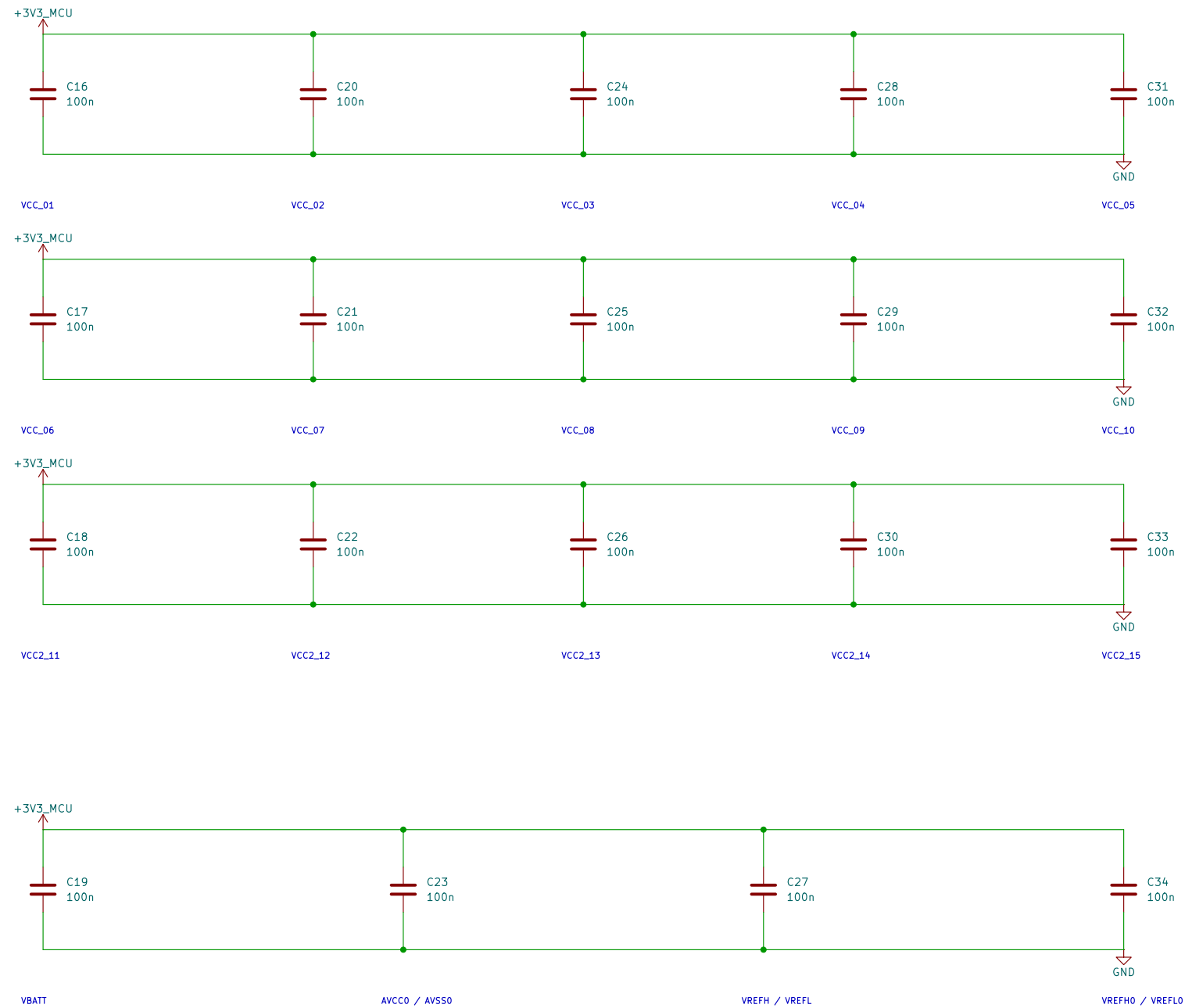
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



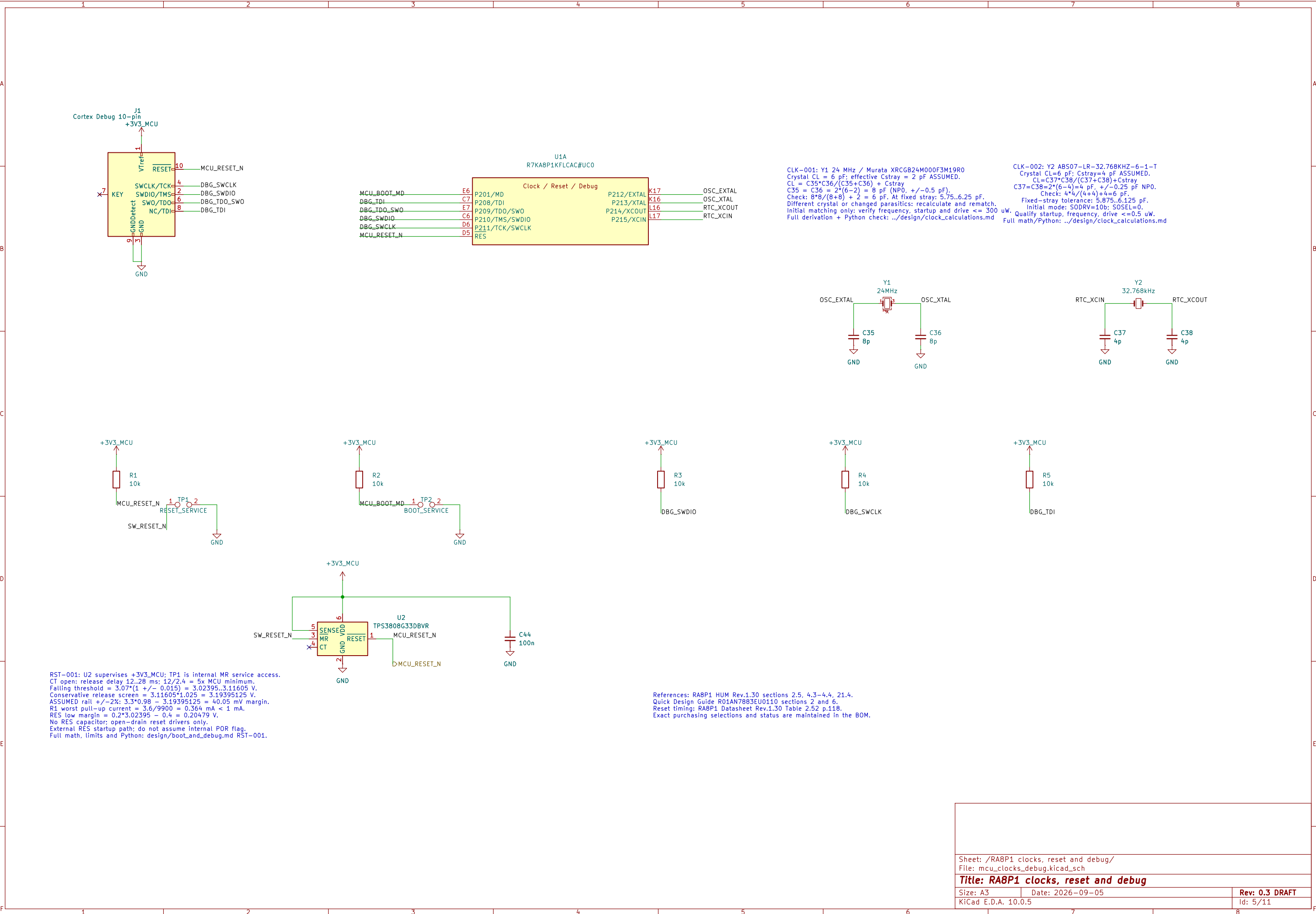


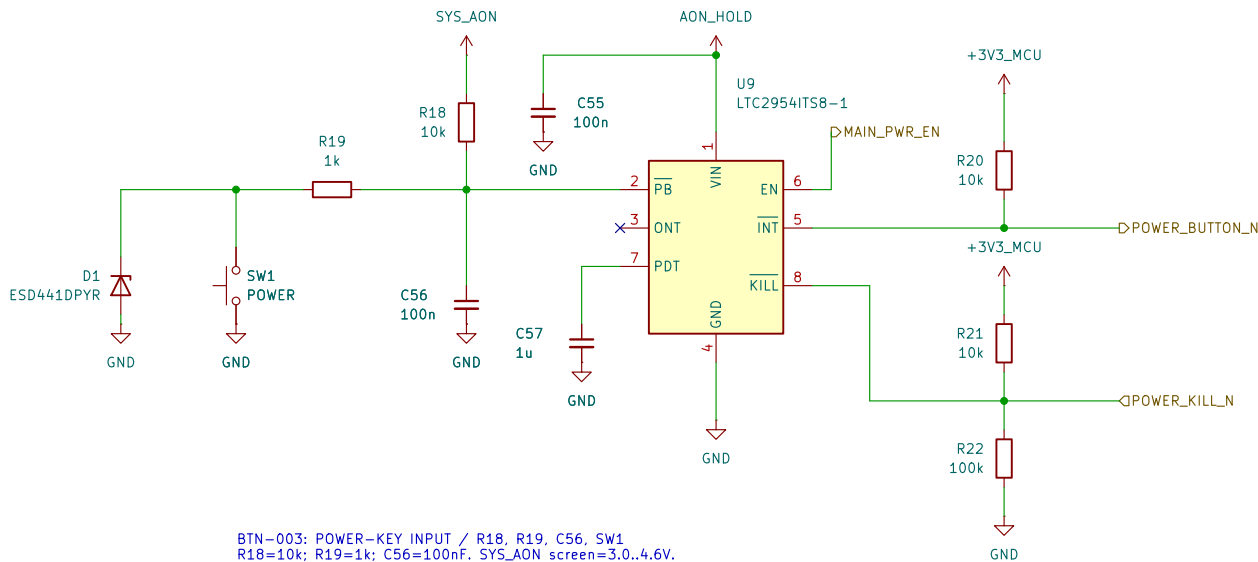


I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



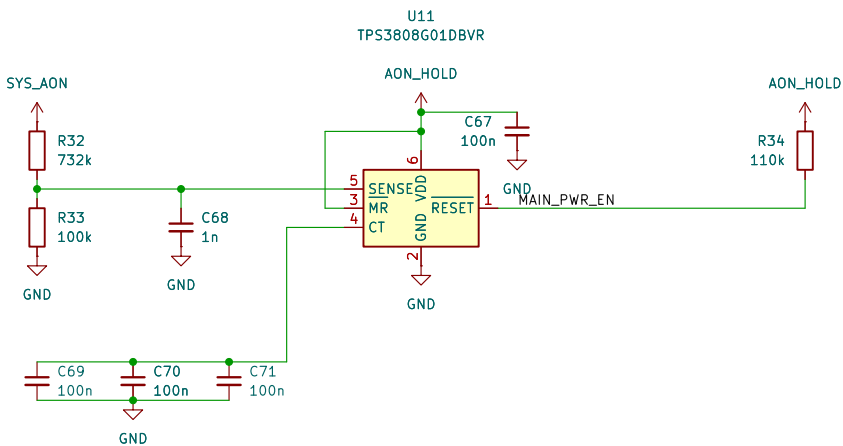
VCC and VCC2 use 3.3 V. All returns use one board ground plane. VBATT follows the main rail; no separate backup-cell supply. Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs. USB/MIP1 supplies remain on the interface sheet and are not yet wired. HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861-862. This is an electrical draft; capacitor MPN qualification remains open.





BTN-003: POWER-KEY INPUT / R18, R19, C56, SW1
R18=10k; R19=1k; C56=100nF. SYS_AON screen=3.0..4.6V.
R bounds: nominal*(0.99*0.99)..nominal*(1.01*1.01).
At PB=0.6V: Isource<=(4.6-0.6)/9801 + 15uA + 12uA
<435.13uA; Isink=0.6/(1020.1+0.5)>587.88uA.
Thus the closed switch pulls PB below its 0.6V falling threshold.
Min contact current=3/(1020.1+1020.1+0.5)-12uA>255.34uA.
Open contact>=2.87748V; SW1 minimum rating=10uA at 2V.
C56 discharge peak<=4.6/980.1<4.694mA < SW1 50mA rating.
12uA is a leakage ALLOCATION; ESD/transient validation required.
Full derivation + Python: ../design/single_button_power.md BTN-003/005/010.

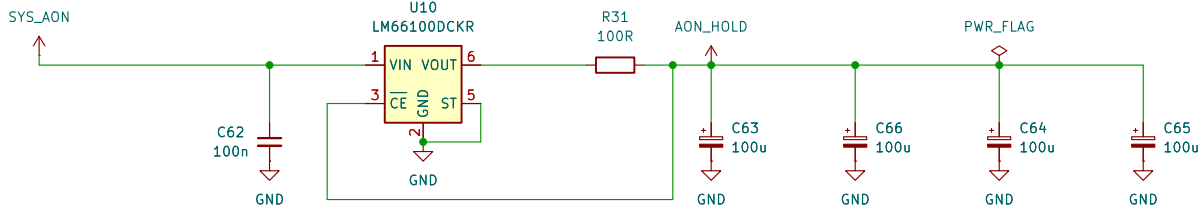
BTN-004: HARDWARE FORCED-OFF TIMING / C57
C57=1uF; TDK C2012X7R1E105K125AB, 25V X7R +/-10%.
C[uF]=1.56e-4*(t_added[ms]-1).
t_force,nom=64ms + 1ms + 1/1.56e-4=6475.256ms (6.48s).
C tolerance/temp only: 1*0.90*0.85=0.765uF;
1*1.10*1.15=1.265uF -> 4.969..8.174s using NOMINAL IC equation.
Not guaranteed timing: excludes IC spread, DC bias, aging, leakage.
ONT intentionally open: internal turn-on debounce=26..41ms.
Release the initial ON press; after forced off, release then press again.
Full calculation + Python: ../design/single_button_power.md BTN-004/005.



SYS-007 / PWR-004: SHUTDOWN / U12, Q1-Q3, R35-R40, R70-R71
POWER_OFF_H=NOT(MAIN_PWR_EN); held-domain VCC=2.7..4.6V.
Each gate branch: 1k series / 1M pulldown; 1% + 100ppm/C, 100C.
4-gate peak<=4*4.6/(1000*0.99^2)=18.773595mA <50mA.
Includes future USB-clear; static<=22.773595uA <100uA.
VGSmin=(2.6-1020.1*1uA)/(1+1020.1/980100)=2.596278V.
1uA per gate/board leakage is a qualification allocation.
RUN/KILL/discharge/USB-clear FET drains remain separate.
R39||R40 + Q2: Rmax=22*1.01^2/2+0.2=11.4211ohm.
Key-filter return allocation Ir=1.5mA; R=Rmax.
Total=10ms+R*C*ln((3.6-Ir*R)/(0.3-Ir*R)).
At 850uF: 34.647839ms; at 1mF: 38.997458ms <46.589403ms.
Each 22R: P<=3.6^2/(22*0.99^2)=0.601052W; 1W at 70C.
Qualify 0.2ohm FET, 10ms TOTAL response, derating and backfeed.
Gate/transient charge shares the existing 1uC reserve.
VBATT firmware/option invariant remains unimplemented; #846 unresolved.
Sources/Python: ../design/system_power_design.md SYS-007;
../design/main_regulator_ltc3119.md PWR-004.

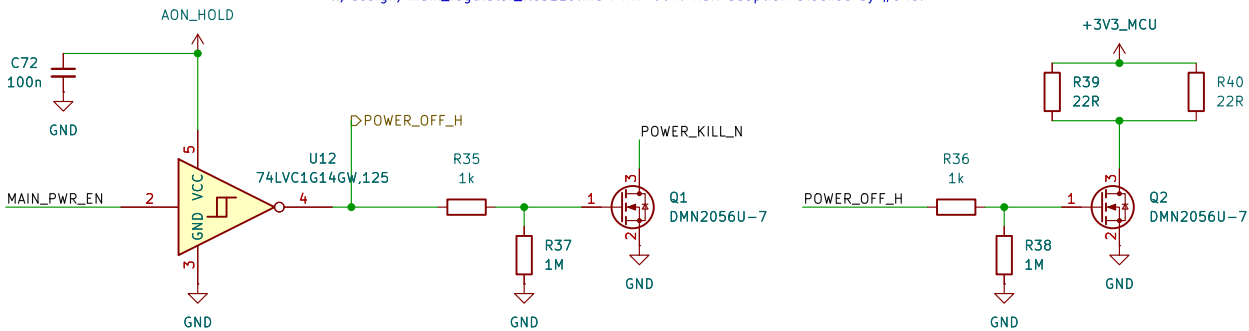
INCOMPLETE POWER CONTROL - NOT FOR FABRICATION
PB, INT, KILL, held supply and source supervisor are connected.
U12/Q1/Q2 provide fault KILL clamp and main-rail discharge.
MAIN_PWR_EN feeds raw U16; held Q3 separately clamps U13 RUN.
SYS_AON source and USB permission-latch clear remain to be built.
Power-control operation is not yet complete or qualified.
Current converter/control draft: ../design/main_regulator_ltc3119.md PWR-004.
Rail adoption is blocked by SDRAM logic-high conflict #846.

ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.



SYS-007 / PWR-004: HELD SUPPLY / U10, R31, C62-C66
C63-C66: 4 x 100uF/10V; Cmin=400uF*0.9^3=291.6uF.
VIN=raw SYS; VOUT->R31->reservoir; CE senses reservoir.
Icontrol<=1.125mA incl 500uA bank leakage allocation.
Pre-trip reverse: 80mV/98.01ohm<0.817mA. Itotal=1.942mA.
Vinitial=3.263705831-max(0.250,0.001125*102.24)
=3.013705831V (threshold OR resistive-drop limit).
t_hold=[291.6uF*(3.013705831-2.7)V-1uC]/1.942mA
=46.589403ms >38.997458ms shutdown at 1mF.
Key-return-aware hold margin=7.591945ms at 1mF.
1uC covers initial undershoot + reverse/gate transitions.
Qualify VIN=0..4.6V, slow/floating-source collapse and load.
R31 short: 4.6^2/98.01<0.216W; 1W nominal rating.
Control only: no MCU/audio/display loads on AON_HOLD.
Separate unadopted eFuse extra load is not included.
Sources/Python: ../design/system_power_design.md SYS-007;
../design/main_regulator_ltc3119.md PWR-004.

SYS-007 / PWR-004: SOURCE SUPERVISION / U11, R32-R34, C67-C71
Vtrip=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/-2%, Isense +/-25nA; each R +/-0.1%, 25ppm/C, 100C.
Corner trip=3.263705831..3.476597632V.
C68: tau=(732k||100k)*1n=87.980769us (nominal).
C69-C71: 3x100n C0G; initial/temp=284.145..316.827nF.
CT delay=300/175+0.0005=1.714786s nominal.
0.6*(284.145/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=0.90s >650ms KILL blank +10ms.
R34=110k; ENhigh>=2.7-3.8uA*110k*1.01^2=2.273598V.
POR sink=(1.3-0.2)/(110k*0.99^2)+3.8uA=14.003041uA.
MAIN_PWR_EN drives raw U16 buffer, not LTC3119 RUN directly.
Held U12 drives KILL/discharge/Q3 RUN clamp; USB-clear still needed.
RUN network and correlated raw/hold threshold screen: PWR-004.
Sources/Python: ../design/system_power_design.md SYS-007;
../design/main_regulator_ltc3119.md PWR-004. Rail adoption blocked by #846.



Sheet: /Power button and source control/
File: power_button.kicad_sch

Title: RA8P1 E-reader - power button and source control

Size: A3

Date: 2026-09-07

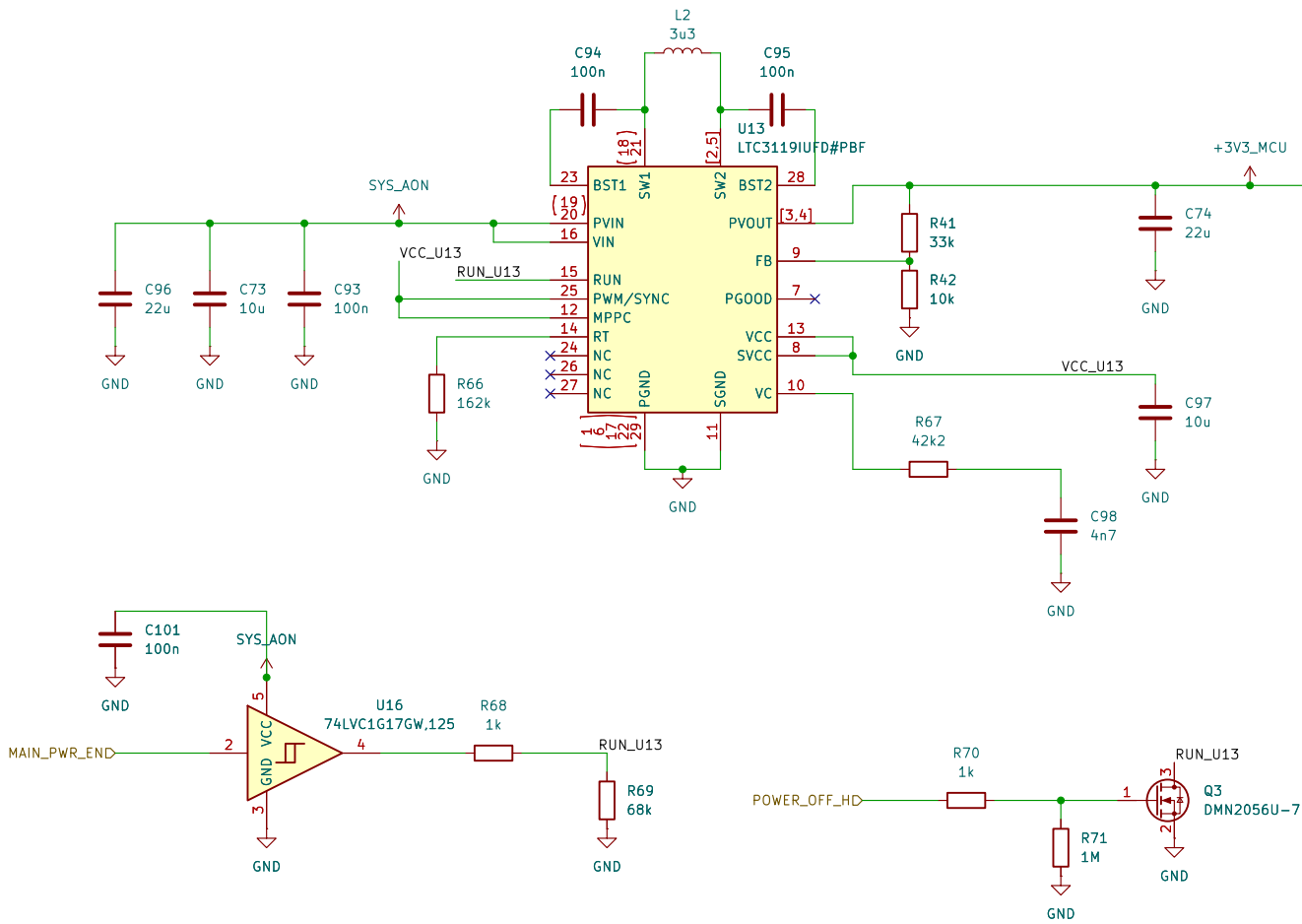
Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.5

Id: 7/11



MAIN DIGITAL SUPPLY: LTC3119 DRAFT / PWR-004
RUN buffer and held shutdown clamp connected.
3.4185V proposal NOT ADOPTED: SDRAM logic margin fails (#846).
Source protection and whole-product verification remain incomplete.
NOT FOR POWER-UP OR FABRICATION.



PWR-004 RUN CONTROL: U16/C101, R68-R71, Q3; R34=110k.
U16 VCC=raw SYS_AON; A=MAIN_PWR_EN; Y->R68 1k->U13 RUN.
R69=68k RUN-to-GND; Q3 drain=RUN after R68; source=GND.
Q3 gate=POWER_OFF_H via R70 1k; R71=1M gate-to-GND.
RUN/KILL/discharge/USB-clear drains remain separate.
Resistor screen: rmin=0.99*2; rmax=1.01*2.
MAIN_PWR_EN adverse leakage allocation=3.8uA.
POR sink=(1.3-0.2)/(110k*rmin)+3.8uA
=14.003041uA <15uA test condition.
ENhigh at held 2.7V=2.7-3.8uA*110k*rmax=2.273598V.
At raw 3.263705831V, held drop<=0.250V; ENhigh>=2.587304V.
Correlate raw/held states with buffer thresholds; interpolation is a screen.
RUNhigh=(3.2-0.1-1uA*1k*rmax)/(1+1k*rmax/(68k*rmin))
=3.052262V.
RUNlow<=0.1+1uA*1k*rmax=0.101020V.
Buffer DC load<=4.6/(68k*rmin)+1uA=70.021uA <100uA.
At raw=0: RUN<=(2uA loff+1uA allocation)*68k*rmax=0.208100V.
This endpoint does not cover undefined buffer behavior below 1.65V.
Held Q3 clamp supplies shutdown during raw collapse.

Four gate branches include the future USB-clear branch:
lpeak<=4*4.6/(1k*rmin)=18.773595mA <50mA absolute limit.
ldc<=4*(4.6/(1M*rmin)+1uA)=22.773595uA <100uA.
VGSmin=(2.6-1uA*1k*rmax)/(1+1k*rmax/(1M*rmin))
=2.596278V.
With installed FET R<=0.2ohm: RUN<=4.6*0.2/(1k*rmin+0.2)
=0.938488mV.
Held-control spare=1125-500-500-4-22.773595=98.226405uA.
1uA leakage and 0.2ohm are qualification allocations, not hot guarantees.
Retain shared 10ms total response and 1uC transition reserve.
Separate proposed eFuse MR/FLT load is NOT included.
Saved draft only: #846 rail/SDRAM conflict remains unresolved.
Full sources, assumptions and Python: ../design/main_regulator_ltc3119.md.

PWR-004: SETPOINT AND COMPENSATION
U13 LTC3119IUFD#PBF; fixed PWM; MPPC disabled via VCC.
R41=33k; R42=10k; each 0.1%, 25ppm/C.
Vnom=0.795*(1+33k/10k)=3.418500V.
For 100C resistor excursion:
fLo=0.999*0.9975=0.9965025; fHi=1.001*1.0025=1.0035025.
Vlo=0.779*(1+33k*fLo/(10k*fHi))-100nA*33k*fHi
=3.328456349V.
Vhi=0.811*(1+33k*fHi/(10k*fLo))+100nA*33k*fHi
=3.509411411V.
100nA is total adverse FB-current ALLOCATION, not a hot guarantee.
Total remaining regulation/routing/ripple/transient allowance=+/-75mV.
Full acceptance envelope=3.253456349-3.584411411V.
MCU release margin=3.253456349-3.193951250=59.505099mV.
Radio margin=3.253456349-0.0875-3.139621574=26.334775mV.
3.6V ceiling margin=3.6-3.584411411=15.588589mV.
These are acceptance budgets, not measured or guaranteed waveforms.

L2=3.3uH +/-20%; qualify installed 2.4uH and hot current.
R66=162k; f=100MHz/(8+1.2*162)=494.071kHz.
Resistor-only corners=484.713, 503.699kHz; silicon variation extra.
R67=42.2k in SERIES with C98=4.7nF COG from VC to GND.
Reduced model, 2160 corners: fc=1,417, 9.966kHz; PM>=51.167deg.
Maximum modeled 0.5A-step excursion=66.112mV.
Model omits distributed PDN, parasitics, delay and saturation.
Installed loop-gain, startup, thermal and transient tests REQUIRED.

Full equations, primary sources and executable Python:
../design/main_regulator_ltc3119.md PWR-004.

PWR-004 DRAFT: LOAD, CAPACITANCE AND SHUTDOWN LIMITS
Main allocation=0.750 MCU+0.500 radio+0.250 SDRAM
+0.250 NOR+0.050 logic=1.800A (not measured maxima).
Audio, SD, camera, display and lighting require separate supplies.
At full Vhi=3.584411411V, Vin=3.2V and ASSUMED eta=0.75:
Pout=Vhi*1.8=6.451941W; Pin=Pout/eta=8.602587W.
lin=Pin/Vin=2.688309A; total stage loss=Pin-Pout=2.150647W.
2.25A cold-load allocation: Pin=10.753234W; lin=3.360386A.
3A temporary screen: Pin=14.337646W; lin=4.480514A.
3A is NOT a qualified continuous rating or source permission.
Radio OFF at cold start; 1.88A reference subtotal excludes extra charging.
Qualify actual loaded startup/ramp, current limit and hot enclosed losses.

Cold storage=147.21u direct+10u C43/FB1+300u C99/C100=457.21uF.
Key filters add 0.4uF through resistors, not parallel loop capacitance.
Partial lower C=110.21u*0.9*0.85*0.6+47u*0.8+300u*0.8*0.8
=280.18639uF; residual factors are ALLOCATIONS, not guarantees.
Qualify 250.850uF loop window; total shutdown storage <=1mF.
C97 internal VCC and input/bootstrap bypasses are NOT main-output C.
C99/C100 initial leakage allowance=2*94.5uA=189uA.

SYS-007 Rdis,max=11.4211ohm; key-return allowance lr=1.5mA.
tfall=Rdis*C*ln((3.6-lr*Rdis)/(0.3-lr*Rdis))+10ms.
At 850uF: 34.647839ms; at 1mF: 38.997458ms.
Existing held-budget screen=46.589403ms; validate all rail/interface tails.
VBATT tied to VCC requires the no-switch initialization/option contract.
Enabled-VBATT 1ms/V fall requirement FAILS the retained-load screen.
Required firmware/option invariant is NOT implemented; see linked record.
RUN buffer and held clamp connected; source protection NOT implemented.
Rail adoption blocked by SDRAM logic-high conflict #846.
No power-up or fabrication release at this checkpoint.

Full sources, assumptions and executable Python:
../design/main_regulator_ltc3119.md PWR-004; SYS-007/009 interfaces.

Sheet: /Main 3V3 digital supply/
File: main_3v3_supply.kicad_sch

Title: RA8P1 E-reader - main digital 3.3 V supply

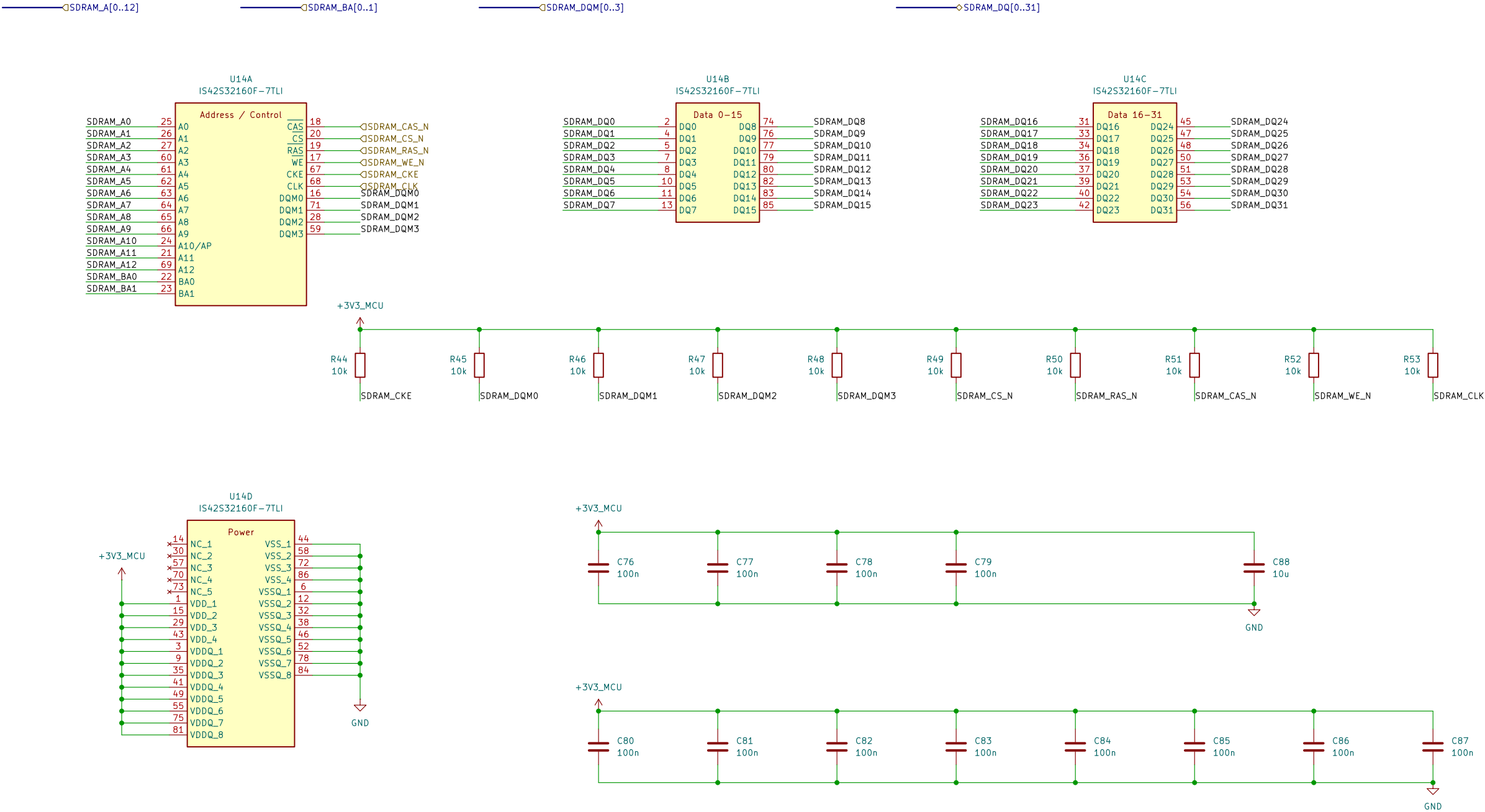
Size: A3 Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.5

Id: 9/11

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

